

Requirements Statement

AI Coding Assistant Tools for Treasury

1 OVERVIEW

The U.S. Department of the Treasury (Treasury), Office of the Chief Information Officer (OCIO), is seeking qualified providers of state-of-the-art Artificial Intelligence (AI)-powered coding assistant tools (similar to GitHub Copilot, AWS CodeWhisperer, Cursor, etc.) that can be delivered as a shared service across the Department and its bureaus.

The intent is to provide developers and IT staff within Treasury with flexible, secure and modern AI-driven coding solutions tailored to their specific development workflows. Selected solutions must operate exclusively on a usage-based pricing model (e.g., per-token, per-completion, or per-query usage). Per-seat licensing or fixed monthly user fees will not be accepted.

2 SCOPE

The requirement covers the procurement of secure, reliable, FedRAMP-authorized, AI-based coding assistant tools to support Treasury's software development lifecycle. The tools must improve productivity, code quality, and speed of delivery across a broad set of environments and use cases, while meeting stringent federal security and compliance mandates.

3 OBJECTIVES

A. General Requirements

The vendor shall provide tools that can meet the following requirements:

1. AI Coding Assistant Capabilities

- Real-time, context-aware code suggestions, completions, and generation.
- Natural-language-driven code generation, editing and explanation
- Support for code review, documentation generation, refactoring, optimization, and test case creation.
- Support for a wide range of programming languages, including but not limited to:
 - TypeScript, JavaScript, Python, Java, C#, C++, Go, SQL, Bash, and PowerShell.

2. Advanced Model Capabilities

- Models comparable in performance to GPT-4, AWS CodeWhisperer, Claude, or similar state of the art large language models (LLMs).
- Demonstrated accuracy and productivity impact in both public and private sector development environments.

- Large-language model (LLM) powered conversational generative AI service.

3. Model Bias and Explainability

- Models mitigate bias and provide explainability of outputs.
- Systems provide transparency into how recommendations are generated and offer user-level control to accept, reject, or customize outputs.

4. FedRAMP Compliance

- FedRAMP Moderate or High impact level authorization (or commitment to obtain within six months after award).

B. Deployment and Integration

- Cloud-based deployment within FedRAMP-authorized Government Cloud, such as
 - Azure GovCloud, AWS GovCloud, Google Cloud Assured Workloads, IBM Cloud for Government.
- Seamless integration with common Integrated Development Environments (IDEs), including
 - Visual Studio Code, JetBrains IDEs, Visual Studio, Eclipse, etc
- Compatibility with major source control and DevOps platforms, including:
 - GitHub, Azure DevOps, GitLab, Bitbucket.

C. Security, Privacy, and Compliance

- Full compliance with federal security standards, including:
 - FISMA Moderate or High, NIST 800-53, and Treasury-specific information assurance requirements.
- US data sovereignty requirements must be met, including:
 - All data processing and storage must occur within U.S.-based data centers, managed and operated by U.S. persons.
- Solutions must guarantee:
 - Treasury data, including source code, prompts, and metadata, will not be used to train or fine tune AI models without explicit written permission from appropriate Treasury personnel (leadership?).

D. Pricing Model

- Solutions must be offered exclusively under a usage-based pricing model, such as:
 - per-token, per-query, or per-completion charges.
- Vendors must include:
 - Transparent and detailed pricing structure
 - Any applicable volume discounts tiers and thresholds
 - Clarity on metering and billing mechanisms (e.g. monthly usage reports)

E. Reliability, Support, and Availability

- Minimum 99.9% uptime Service Level Agreement (SLA) must be provided.

- Detailed support framework must be included setting forth:
 - Support hours (24x7 availability)
 - Incident response and escalation procedures
 - Designated support contacts or account representatives
- Vendor should indicate:
 - Availability of self-service resources such as knowledge bases, FAQ's, and developer documents

F. Training, Onboarding, and Change Management

- Vendors shall provide:
 - Training resources, onboarding support, and user enablement strategies.
 - Instructional materials (videos, tutorials, workshops) offered to accelerate adoption.

G. Vendor Viability and Roadmap Alignment

- Vendors must provide:
 - Long-term viability, financial stability, and ability to support Treasury over the full solution lifecycle.
 - Alignment of their product roadmap with evolving federal AI policy, security expectations, and developer needs.